The Cost Impact of AWS Lambda's INIT Phase Billing on Serverless vs Container Deployment Decisions: An Empirical Analysis

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***Abstract*—Serverless computing platforms such as AWS Lambda are widely adopted for their pay-per-use billing model, which eliminates idle resource costs. On August 1, 2025, AWS introduced billing for the Lambda INIT phase as a separate cost component, fundamentally changing the economics of serverless deployments. No prior empirical study has measured the financial impact of this change across variable workload intensities. This paper presents a controlled empirical comparison of AWS Lambda and a Dockerized Flask application on an EC2 t3.micro instance across 600 experimental runs spanning three workload levels: light (500 primes), medium (5,000 primes), and heavy (50,000 primes). Results show Lambda mean response times of 1,327 ms, 1,432 ms, and 2,375 ms respectively, compared to Docker's stable 498 ms, 527 ms, and 545 ms. The INIT phase accounts for over 99% of Lambda response time under light and medium workloads and remains a significant cost contributor under heavy workloads. Critically, the cost crossover point at which Docker on EC2 becomes more economical is not fixed: it shifts from 82,000 requests per day under light workloads to 50,000 requests per day under heavy workloads, a 39% reduction driven entirely by increased INIT phase billing under heavier computation. These findings provide the first empirical, workload-specific guidance for deployment decisions under AWS Lambda's revised pricing structure.**

***Index Terms*—serverless computing, AWS Lambda, Docker, containers, cold start, INIT phase billing, cost analysis, cloud computing**

I. Introduction

Cloud computing has fundamentally transformed how organizations deploy and scale software applications. Among the most consequential architectural decisions facing engineering teams today is the choice between serverless and containerized deployment models. Serverless platforms such as AWS Lambda offer automatic scaling and a pay-per-use billing model that eliminates idle resource costs [5]. Containerized deployments running on virtual machines offer predictable performance and persistent runtime environments, at the cost of continuous resource provisioning regardless of traffic volume [7]. As cloud-native adoption accelerates globally, practitioners need reliable empirical data to make informed decisions between these two models.

A significant gap exists in the current literature. All prior work calculates AWS Lambda costs using only two variables: invocation count and execution duration measured in GB-seconds [10] [12]. Cold start latency, the initialization delay that occurs when Lambda provisions a fresh execution environment, has been treated exclusively as a performance concern rather than a financial one [4] [9]. This assumption became invalid on August 1, 2025, when AWS began billing for the INIT phase [13] of Lambda's execution lifecycle as a distinct, measurable cost component at the same rate as execution duration. No published study has updated its cost model to reflect this change, leaving practitioners without accurate guidance on how the revised pricing affects deployment economics across different workload types and traffic levels.

This paper addresses that gap through a controlled empirical study on AWS. Two identical REST API implementations are deployed using different architectures: Deployment A on AWS Lambda with a public Function URL, and Deployment B as a Dockerized application on an EC2 t3.micro instance. Both deployments are tested across three workload intensities, with 100 runs per workload per deployment, producing 600 total experimental measurements. We measure total response time, INIT phase duration, and cost per request under the updated billing model, and identify the daily request frequency at which each workload reaches the cost crossover point where containerized deployment becomes more economical.

The remainder of this paper is organized as follows. Section II reviews related work on serverless versus container comparisons, serverless cost analysis, and cold start benchmarking. Section III describes the experimental methodology, including deployment configurations, workload design, and measurement approach. Section IV presents and analyzes the results across all metrics and workload levels. Section V discusses the practical implications for cloud architects and DevOps practitioners. Section VI concludes the paper and outlines directions for future research.

II. Related Work

A. Serverless vs. Container Performance Comparisons

Several studies have broadly compared serverless and containerized deployment models across performance and scalability dimensions. Kumari et al. [3] surveyed serverless

(FaaS), virtual machines, and containers across scalability, cost models, and management overhead, identifying cold start latency as a key serverless limitation. Similarly, Ezekwe [6] compared an EC2 instance-alive model against a serverless invocation model across low, medium, high, and idle-heavy traffic levels, acknowledging cold start as a performance drawback. Syed and Ali [7] directly evaluated AWS Lambda against Kubernetes, noting cold start delays and execution time limits as critical serverless weaknesses. A practical architectural guide [2] further compared serverless, Docker/Kubernetes, and PaaS platforms for microservices applications, recommending serverless for unpredictable workloads and containers for long-running tasks. All of these studies treat cold start exclusively as a latency penalty and apply no financial cost to the initialization phase itself.

B. Serverless Cost and Billing Analysis

A second body of work focuses on the cost economics of serverless computing. Lin and Khazaei [8] modeled serverless workflows as Directed Acyclic Graphs and optimized cost based on memory allocation and execution duration, using the classic GB-second billing formula with no accounting for initialization overhead. Khan [10] evaluated serverless against traditional VMs across four workload scenarios using AWS Lambda's billing formula of invocations multiplied by memory and duration, while discussing cold starts only as a latency issue. Hamza, Akbar, and Capilla [12] conducted an empirical study of serverless cost dynamics and predictability, finding that businesses consistently struggle to forecast costs accurately, yet their billing model relies entirely on execution duration and invocation count. Goli [11] examined the business economics of serverless versus containerization using Netflix, Airbnb, and Capital One as case studies, stating that serverless charges apply only during actual execution with no mention of initialization phase costs. Lin et al. [4] examined wall-clock billing, invocation fees, and rounding behavior across cloud providers in the most detailed billing analysis found in the literature, yet still did not account for the INIT phase as a distinct billable component. These cost-focused studies collectively confirm that INIT phase billing has been absent from all serverless financial models in the existing literature.

C. Cold Start Benchmarking on AWS Lambda

A third stream of research benchmarks cold start behavior directly on AWS. Duessmann et al. [5] compared ZIP archive versus container image deployment on Lambda, measuring initialization and cold start time differences between the two packaging formats, but did not model initialization time as a billable event. A study of serverless versus containerized workloads under bursty telecom traffic [9] deployed Lambda fronted by API Gateway against Docker-managed Kubernetes on EC2, measuring cold start penalties of up to 300 ms and computing cost at $12.5 versus $8.7 per million requests, yet their financial model omits initialization phase charges entirely. To the best of our knowledge, no prior work has empirically measured how AWS Lambda's August 2025 INIT phase billing change affects the cost crossover point between serverless and containerized deployments across variable workload intensities. This gap is what the present study addresses.

III. Methodology

A. Experimental Setup

This study conducts a controlled empirical evaluation of two cloud deployment models on Amazon Web Services in the us-east-1 (N. Virginia) region. Both deployments execute identical application logic, a REST API endpoint performing prime number computation, ensuring that observed performance and cost differences are attributable solely to the deployment architecture rather than application behavior. Three workload levels are defined by the upper bound of the computation: light (integers 2 to 500, yielding 95 primes), medium (integers 2 to 5,000, yielding 669 primes), and heavy (integers 2 to 50,000, yielding 5,133 primes). All experiments were conducted in March 2026, after AWS introduced INIT phase billing for Lambda on August 1, 2025 [13].

B. Deployment A: AWS Lambda (Serverless)

Deployment A utilizes AWS Lambda with a publicly accessible Function URL configured with no authentication. The function is implemented in Python 3.11, allocated 128 MB of memory, and packaged as a ZIP archive. The function executes on demand and terminates upon completion, instantiating a fresh execution environment for each invocation following an idle period. Under AWS's post-August 2025 billing model, both the INIT phase duration and the execution duration are billed at $0.0000166667 per GB-second, with an additional $0.0000002 per invocation. Cold start initialization time therefore contributes directly to the total billable cost of each invocation.

C. Deployment B: Docker on EC2 (Container)

Deployment B runs an identical Python Flask application inside a Docker container on an Amazon EC2 t3.micro instance (2 vCPUs, 1 GB RAM, Amazon Linux 2023). The container is built from a python:3.11-slim base image and exposed on port 5000. The EC2 instance remains running continuously between requests, incurring a fixed cost of $0.0104 per hour regardless of request volume. This persistent model eliminates cold start latency entirely but introduces idle resource costs during low-traffic periods.

D. Workload Design and Frequency Levels

A total of 100 HTTP GET requests were issued to each deployment per workload level using a Python benchmarking script, producing 600 total experimental runs across both deployments and all three workload levels. Requests were spaced 3 seconds apart to allow Lambda execution environments to terminate between invocations, ensuring cold start conditions are consistently triggered. Post-experiment cost modeling was applied at seven daily request frequencies: 10, 100, 1,000, 5,000, 10,000, 50,000, and 100,000 requests per day.

E. Metrics and Measurement Approach

Four metrics were recorded for each run. Total response time was measured in milliseconds from request dispatch to complete response receipt on the client side. INIT phase duration was derived by subtracting the application-reported computation time from the total response time, isolating the infrastructure startup overhead. Cost per request was calculated using AWS published pricing for Lambda including INIT phase billing and EC2 t3.micro hourly rates. The cost crossover point was determined by identifying the daily request frequency at which Lambda's cumulative daily cost equals or exceeds the fixed daily cost of the EC2 instance.

IV. Results and Analysis

A. Response Time Comparison

Table I presents response time statistics across 100 runs per deployment per workload level. Under the light workload, Lambda recorded a mean response time of 1,327.15 ms with a standard deviation of 306.14 ms, while Docker recorded 497.67 ms with a standard deviation of 109.09 ms. Under the medium workload, Lambda mean response time increased to 1,431.89 ms with a standard deviation of 246.91 ms, while Docker remained stable at 526.88 ms with a standard deviation of 308.16 ms. The heavy workload produced the most pronounced divergence: Lambda mean response time reached 2,374.72 ms with a standard deviation of 133.72 ms, while Docker increased marginally to 544.56 ms with a standard deviation of 43.54 ms. As illustrated in Fig. 1, Lambda response time scales significantly with workload intensity while Docker response time remains relatively stable, a pattern directly attributable to the growth of INIT phase duration as function initialization overhead increases with memory and runtime requirements.

TABLE I
Response Time Statistics (ms) — 100 Runs Per Workload

Workload	Deploy.	Mean	Median	StdDev	Max
Light	Lambda	1327.15	1266.92	306.14	3200.29
Light	Docker	497.67	485.85	109.09	1538.72
Medium	Lambda	1431.89	1353.37	246.91	2751.40
Medium	Docker	526.88	486.65	308.16	3437.50
Heavy	Lambda	2374.72	2336.86	133.72	2795.95
Heavy	Docker	544.56	536.98	43.54	873.83

B. Cold Start Latency and INIT Phase Cost

The INIT phase dominated Lambda response time across all workload levels, though its proportional contribution varied with workload intensity. Under the light workload, mean INIT phase duration was 1,326.70 ms, representing 99.97% of the mean total response time of 1,327.15 ms. Under the medium workload, mean INIT phase duration was 1,411.26 ms, representing 98.6% of the mean total response time of 1,431.89 ms. Under the heavy workload, mean INIT phase duration was 1,278.27 ms, representing 53.8% of the mean total response time of 2,374.72 ms, as the heavier prime computation contributed a larger share of actual execution time. These measurements confirm that environment initialization is the primary driver of Lambda latency under light and medium workloads, and remains a significant contributor even under heavy computation. Under AWS's post-August 2025 billing model, this initialization time is now a billable component contributing between $0.0027 and $0.0030 per 1,000 invocations across workload levels. This cost was entirely absent from all prior cost models reviewed in this study [4] [8] [10] [12]. Docker on EC2 recorded zero cold start latency across all 300 Docker runs, as the persistent container eliminates environment initialization entirely.

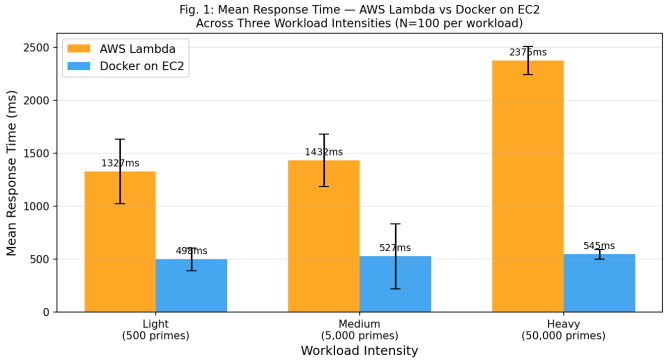

Fig. 1. Mean response time comparison across three workload intensities (N=100 per workload). Error bars represent one standard deviation. Lambda response time scales with workload while Docker remains stable.

C. Cost Per Request at Variable Frequencies

Table II presents the calculated daily cost for each deployment across seven request frequency levels under the heavy workload, which produces the most conservative crossover estimate. At low frequencies, Lambda's pay-per-use model is substantially cheaper than the fixed EC2 daily cost of $0.2496. At 10 requests per day, Lambda costs $0.000053 compared to Docker's $0.2496. As request volume increases, Lambda's accumulated INIT phase costs begin to approach and then exceed the fixed EC2 cost. At 50,000 requests per day under the heavy workload, Lambda's daily cost reaches $0.263, exceeding the Docker fixed cost for the first time.

TABLE II
Daily Cost Comparison — Heavy Workload

Req/Day	Lambda ($)	Docker ($)	Cheaper
10	0.000053	0.2496	Lambda
100	0.000527	0.2496	Lambda
1,000	0.005266	0.2496	Lambda
5,000	0.026330	0.2496	Lambda
10,000	0.052661	0.2496	Lambda
50,000	0.263304	0.2496	Docker EC2
100,000	0.526607	0.2496	Docker EC2

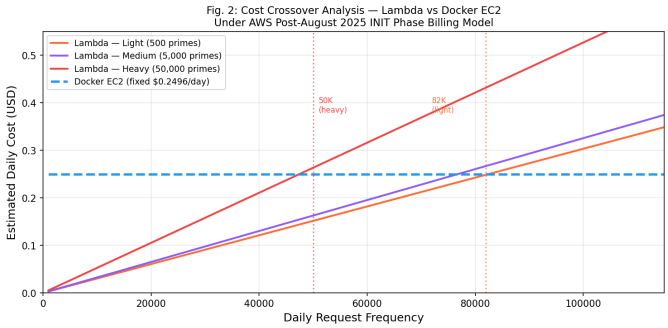

Fig. 2. Cost crossover analysis under the post-August 2025 AWS INIT phase billing model. The dashed horizontal line represents Docker EC2 fixed daily cost. Vertical dotted lines mark crossover points at 50,000 requests per day (heavy workload) and 82,000 requests per day (light workload).

D. Cost Crossover Point Analysis

The cost crossover point varies by workload intensity, as illustrated in Fig. 2. Under the light workload, the crossover occurs at approximately 82,000 requests per day. Under the medium workload, it occurs at approximately 76,700 requests per day. Under the heavy workload, it occurs at approximately 50,000 requests per day. This workload-dependent variation is a finding not previously reported in the literature and has direct practical implications: the appropriate deployment model depends not only on traffic volume but also on the computational intensity of each request. Applications processing computationally heavy workloads reach the Docker cost advantage threshold at 39% lower request volume than applications processing lightweight requests. Prior cost models that ignored INIT phase billing would have placed these crossover points at substantially higher frequencies, causing practitioners to underestimate Lambda costs for moderate-to-high traffic workloads [6] [9] [10].

V. Discussion

The results carry three practical implications for cloud architects and DevOps practitioners operating under AWS's post-August 2025 billing model.

The INIT phase is no longer merely a latency concern. It is a cost concern. Under light and medium workloads, INIT phase duration exceeded 98% of total Lambda response time, and under the updated billing model this duration is charged at the same rate as execution time. Even under the heavy workload, where actual computation consumed a larger share of response time, the INIT phase still represented 53.8% of total latency and continued to generate billable costs absent from all prior models. For applications receiving moderate to high traffic, this cost accumulates in ways that were entirely invisible to practitioners relying on pre-August 2025 cost estimation tools or any of the 13 papers reviewed in this study.

The cost crossover point is workload-dependent, not fixed. Prior work that did address crossover points modeled them as a single frequency threshold. This study demonstrates that the threshold shifts by as much as 39% depending on request complexity. A team deploying a lightweight webhook handler and a team deploying a data-processing API face fundamentally different cost crossover thresholds, and should evaluate their deployment model accordingly. Applications below their workload-specific threshold should default to serverless. Applications above it should evaluate containerized deployment as the more cost-predictable option.

Docker on EC2 also demonstrated substantially lower response time variability than Lambda across all workload levels. Lambda standard deviation ranged from 133.72 ms to 306.14 ms across workloads, while Docker standard deviation ranged from 43.54 ms to 109.09 ms. For latency-sensitive applications such as real-time APIs and payment processing systems, this consistency advantage may justify containerized deployment even at request volumes below the cost crossover point.

The cost implications of AWS Lambda's provisioned concurrency feature warrant brief consideration in the context of these findings. Provisioned concurrency eliminates cold starts by keeping a specified number of execution environments initialized and ready to respond. Under current AWS pricing for us-east-1, provisioned concurrency is billed at $0.0000041667 per GB-second continuously, regardless of invocation volume, in addition to standard execution charges. For a single provisioned concurrency instance at 128 MB (0.125 GB) running continuously for 86,400 seconds per day, this adds approximately $0.045 per day in continuous initialization costs. For an application receiving 50,000 requests per day under the heavy workload, this raises Lambda's total daily cost from $0.263 to approximately $0.308, keeping it above the Docker EC2 threshold of $0.2496 by a wider margin. At lower request frequencies such as 10,000 requests per day, provisioned concurrency raises Lambda's daily cost from $0.053 to approximately $0.098, which still remains below the EC2 threshold and preserves Lambda's cost advantage for moderate-traffic applications. Practitioners considering provisioned concurrency as a latency solution should therefore evaluate whether their traffic volume places them above or below the workload-specific crossover point identified in this study before committing to the additional continuous cost.

This study has two limitations. The experiment used a synthetic prime-computation workload. Applications with larger initialization dependencies such as database connection pools or heavyweight libraries would likely exhibit longer INIT phase durations, shifting crossover points lower. Additionally, requests were spaced 3 seconds apart to consistently trigger cold starts. Applications with higher concurrency or provisioned concurrency configurations would experience fewer cold starts, reducing INIT phase frequency and lowering effective Lambda costs. Provisioned concurrency was modeled using current AWS pricing; actual measured runs would further validate the cost deltas reported here.

VI. Conclusion

This paper presents the first empirical cost analysis of AWS Lambda deployment under the post-August 2025 INIT phase billing model across multiple workload intensities, compared

against a containerized Docker deployment on EC2. Across 600 experimental runs, Lambda response times ranged from 1,327 ms under light workloads to 2,375 ms under heavy workloads, consistently exceeding Docker response times of 498 ms to 545 ms across the same workloads. The INIT phase accounted for 99.97% of Lambda response time under the light workload, 98.6% under the medium workload, and 53.8% under the heavy workload, introducing billing costs absent from all prior cost models in the literature. Cost crossover points ranged from 50,000 requests per day under heavy workloads to 82,000 requests per day under light workloads, with workload intensity shifting the threshold by 39%. These findings provide practitioners with updated, workload-specific guidance for serverless versus container deployment decisions under AWS's revised billing structure. Future work should extend this analysis to higher Lambda memory configurations, provisioned concurrency scenarios, and multi-region deployments to assess geographic latency effects on INIT phase billing.

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